

Computer Performance Evaluation (Spring 2026)

Section	Lecture Time and Room	Department
CSC 641/841-01	Tuesday 4:00-6:45PM HSS 362	Computer Science

Instructor: Joseph Hui, jhui@sfsu.edu

Course Website: All of the course materials will be available through Canvas. <http://canvas.sfsu.edu>

Office Hours: By appointment only (email and I'll schedule a time. Zoom link on Canvas).

CSC 641/841 is a paired course for undergraduate and graduate students. Undergrad students must enroll in CSC 641 and graduate students must enroll in CSC 841.

Prerequisites:

- CSC 641: CSC 415 with a grade of C or better.
- CSC 841: graduate standing or consent of the instructor.
- Students who have completed CSC 641 may not take CSC 841 for credit.
- The prerequisites for this course are strictly enforced for all sections. Concurrent enrollment of CSC 415 is not allowed.

TextBook: *Systems Performance – Enterprise and the Cloud*, Second Ed., by Brendan Gregg

Recommended Reading: *Art of Computer Performance Analysis – Techniques for Experimental Design Measurements Simulation and Modeling*, by Raj Jain

Objectives: Computer performance analysis problems related to system design, selection, and tuning. Modeling using stochastic and operational queuing models. Workload characterization, design, and performance measurement methods. Design of simulation models for computer systems.

Tentative Course Outline:

- Performance Terminology and Methodology
- Benchmarking
- Observability
- Performance tuning

Grading Policy:

1. Four Milestones 50% (12.5% each)
 - a. Milestone 1 – Install Linux Container - generate output for top, vmstat, iostat, nstat, perf
 - b. Milestone 2 – Benchmark – O/S level - 10 sec run versus 10,000 iterations
 - c. Milestone 3 – Benchmark – Application - 10 sec run versus 10,000 iterations
 - d. Milestone 4 – Multi-tier Performance Analysis
2. Midterm Exam I: 25%
3. Midterm Exam II: 25%
4. Final Exam (optional): 25% – best two (2) scores from Midterm 1, Midterm 2, and Final

Tentative Schedule:

Week	Date	Pre-reading Assignment	Topic	Assignments
1	1/27	Chapter 3	Syllabus and Course Policies Intro to Performance Analysis	Milestone 1 assigned
2	2/3	Chapter 1	Performance Terminology	
3	2/10	Chapter 2	Performance Methodology	
4	2/17	Chapter 4, 6, 7	Observability Tools Part 1 CPU & Memory	Milestone 1 due Milestone 2 assigned
5	2/24	Chapter 8, 10	Observability Tools Part 2 File Systems & Network	
6	3/3		Midterm 1	
7	3/10	Chapter 12	Benchmarking & Benchmarking Methodology	
8	3/17	Chapter 5	Performance Analysis of Applications & Multi-Tier Applications	Milestone 2 due Milestone 3 assigned
9	3/24		NO CLASS	
10	3/31		NO CLASS	
11	4/7		Quality Assurance (QA) - Unit, Regression, Benchmark testing & Deployment strategies/tools	Milestone 3 due Milestone 4 assigned
12	4/14		Code Performance Issues & Writing Performant Code	
13	4/21		RDBMS Performance Analysis and Tuning	
14	4/28		Midterm 2	
15	5/5	Chapter 16	Case Studies Sephora and Oracle	Milestone 4 due
16	5/12		Guest Lecture/Final Review	
17	5/19		FINAL	

Communication:

- Email is the best way to communicate with me outside of class.
- Feel free to send me questions about the software industry or your resume for feedback.
- This is a live document; I may make changes to any of these policies and contents at any time during the semester. I will announce changes in class and post them on Canvas.

Important Dates:

1. Midterm I Exam:
 - a. Tuesday, March 3, 4:00-6:00pm
2. Midterm II Exam
 - a. Tuesday, April 28, 4:00-6:00pm
3. Final Exam:
 - a. Tuesday, May 19, 5:00-7:00pm

Deadlines: All the assignments will be due on Tuesdays at 4:00pm. Please read the course policies to learn more about the policies for checkpoint assignments in this course.

Grades: The final letter grade for the course will be computed using the following distribution:

Highest	Lowest	Letter Grade
100	93	A
92.99	90	A-
89.99	87	B+
86.99	83	B
82.99	80	B-
79.99	77	C+
76.00	73	C
72.99	70	C-
69.99	67	D+
66.99	63	D
62.99	60	D-
59.99	0	F

Course Policy:

- **Attendance:** Students are expected to attend all scheduled lectures in this course. Attendance is vital for understanding course material, participating in class discussions, and achieving academic success. Students are responsible for what they miss if they don't attend any of the scheduled lectures
- **Exams:** All the exams in this course will be given on paper, in the classroom and will be proctored by the instructor. A government photo ID (student ID or driver's license) is required when turning in exam. If any of the scheduled exam dates conflict with your religious observances, you must notify your instructor during the first two weeks of the semester. Drop and withdrawal from courses are student responsibilities. Since two best scores out of three test scores are used to determine the final grade, no makeup exams are allowed (i.e., you can miss a Midterm but will then need to take the final).
- **Milestones Assignments:** This class has five mandatory milestones assignments. The following are their policies:
 - Only assignments uploaded to GitHub or Canvas will be considered for grading.
 - All assignments are due on the designated due date, as specified by the instructor, and it is the student's responsibility to keep track of all assignment due dates and plan accordingly.
 - Students are responsible for what they turn in. If you submit a blank assignment, it will be graded as such. Always double check your submissions to make sure what you meant to upload is actually what you uploaded.
 - Late homework submissions will be subject to penalties. A 10% deduction will be applied for each day that the assignment is overdue, up to a maximum of three days. After three days, the assignment will be considered as not submitted and will be graded as such. This policy applies only to milestones 1, 2, 3 and 4. Milestone 5 must be submitted on time because it is the last milestone of the semester. It is imperative for students to adhere to the specified deadlines, as failing to do so will result in a reduction of their assignment scores. This policy is designed to emphasize the importance of timely and responsible submission of coursework, while also providing a reasonable window for unforeseen circumstances. Students are strongly encouraged to manage their time effectively to avoid any unnecessary deductions and to ensure the best possible academic outcomes.
- **Class Notes:** Students may not capture audio, photos or video from class sessions on their own devices without the explicit permission of the instructor and everyone present, unless part of a DPRC-authorized accommodation.
- **Office Hours Policies:** Office hours are a valuable resource provided by instructors to support students' learning and offer additional assistance outside of regular lectures. They are intended to foster a deeper understanding of course materials, answer specific questions, provide guidance on assignments, and address any concerns related to the course. It is important to emphasize that office hours are not intended to serve as a replacement for missed lectures. Instructors do not typically use office hours to re-lecture or repeat the content covered during a missed class.
- **Instructor Availability:** I will be unable to respond to emails or messages during weekends/holidays. You can still send an email or message asking for help, but I will be unable to provide a response to your request until the next business day
- **Final Grades:** I create a lot of opportunities for students to do well in this course, and all posted grades are final unless there was a verifiable mistake in the grade calculation. Fairness demands, and university rules, require that all students be marked according to the same standards, so that all students receive the marks they earned. I can't just raise your final grade for unjustifiable reasons, so I will not respond to any such requests.

- **Academic Integrity:** Any form of cheating or plagiarism will incur very serious consequences. Students may not post any course materials to any third-party sites (such as Chegg) or post any recordings, screenshots, audio or chat transcripts in any setting outside the class, and that violations of this are subject to student disciplinary action. Any instances of cheating, deceit, fabrication, forgery, plagiarism, unauthorized altering of records or submitting false documents, unauthorized collaboration, unauthorized submission of work previously given credit, or other forms of academic misconduct will be assigned a grade penalty, likely an F or a grade of zero. Failing one or more assignments or examinations for reasons of academic integrity violations may result in a final class grade of F. Students may not withdraw from classes in which they have committed academic misconduct. Consequences for violations of academic integrity may exceed an F on the assignment, examination, or class as determined by the Academic Integrity Review Committee. Members of our academic community have a responsibility to develop an awareness of academic integrity, to cultivate skills to realize honesty in academic and community work, and to sustain actively academic honor as a core value of our community. Students are expected to engage in behaviors that reflect well upon the university. In addition to attending to one's own actions, the Standards for Student Conduct require that students who witness academic dishonesty notify their faculty/instructor, department chair, or the Office of Student Conduct. Supporting academic integrity enhances the reputation of the University and the value attributed to degrees awarded by the University. To learn more about the university policies regarding academic integrity: <https://conduct.sfsu.edu/academic-dishonesty>.
- **Students with disabilities:** Students with disabilities who need reasonable accommodations are encouraged to contact the instructor. The Disability Programs and Resource Center (DPRC) is available to facilitate the reasonable accommodations process. The DPRC is located in the Student Service Building and can be reached by telephone (voice 415-338-2472, video phone 415-335-7210) or by email (dprc@sfsu.edu).
- **SF State fosters a campus free of sexual violence** including sexual harassment, domestic violence, dating violence, stalking, and/or any form of sex or gender discrimination. If you disclose a personal experience as an SF State student, the course instructor is required to notify the Title IX Coordinator by completing the report form available at <http://titleix.sfsu.edu>, emailing vpsaem@sfsu.edu or calling 338-2032. To disclose any such violence confidentially, contact: The SAFE Place - (415) 338-2208; http://www.sfsu.edu/~safe_plc/ Counseling and Psychological Services Center - (415) 338-2208; <http://psyservs.sfsu.edu/> For more information on your rights and available resources: <http://titleix.sfsu.edu>
- **The Religious and Cultural Holidays policy (S09-212)** states that the faculty of San Francisco State University shall accommodate students wishing to observe religious and cultural holidays when such observances require students to be absent from class activities. It is the responsibility of the student to inform the instructor, in writing, about such holidays during the first two weeks of the class each semester. If such holidays occur during the first two weeks of the semester, the student must notify the instructor, in writing, at least three days before the date that they will be absent. It is the responsibility of the instructor to make every reasonable effort to honor the student request without penalty, and of the student to make up the work missed.
- **Academic calendar policy:** The Academic Calendar can be found at <https://webapps.sfsu.edu/public/webcal/acadcalendar>

Immigration Support and Assert Your Rights

No human being is illegal, and everyone has a right to education. First and foremost, your safety and confidentiality are my priority. I do not and will not condone or enable intimidation (by peers, faculty, staff, local or federal agencies) of anyone based on their documentation status. In our classroom, you are never required to share this information. However, in this time of increased fear and uncertainty, it is important that you know your rights and strengthen your connection with those on/off campus who can support you. I am also available for confidential discussions that help me better support you while we are on campus together.

- [SFSU Dream Resource Center](#)
- [CSU Resources for Undocumented Students](#)
- [Legal support services](#)
- [ACLU Know Your Rights](#) when encountering law enforcement.
- [Know Your Rights cards from the Immigrant Legal Resource Center](#)
- [Mental health resources](#)
- [Immigration Law Help \(search your zip code\)](#)
- [Immigrant Legal Resource Center \(ILRC\)](#) and [National Immigration Law Center \(NILC\)](#)
- [National Immigrant Justice Center: What to do if you or are loved one are detained](#)
- Rapid Response Numbers If You See ICE or Someone Detained by ICE:
- [San Francisco Immigrant Response Help](#) (415) 200-1548
- [Alameda County Immigration, Legal, and Education Partnership](#) (510) 241-4011
- [San Mateo County Rapid Response Hotline](#) (203) 666-4472 / (203) NO-MIGRA
- [Santa Clara County Rapid Response Network](#) (408) 290-1144
- [Stand Together Contra Costa County](#) (925) 900-5151

Your designated contact on campus for concerns about immigration orders and inquiries is **Jamillah Moore**, Vice President for Student Affairs & Enrollment Management, or **Frederick Smith**, Associate Vice President for Equity, Community, and Inclusion.

They can be reached at (415) 338-2032 or vpsaem@sfsu.edu